Data Cleaning Change Log

DecodeLabs | Data Analytics | Project 1: Data Cleaning & Preparation
Prepared by: Sanjana Kammar | Date: 17 October 2026 | Tool: Python (pandas)
Dataset: Dataset_for_Data_Analytics.xlsx (1,200 rows, 14 columns, e-commerce orders). Cleaned output: cleaned_dataset.csv. Reproducible script: clean_data.py.

1. Audit summary (before cleaning)

Check	Result
Rows / columns	1,200 / 14
Missing values	309 blanks, all in CouponCode. No other column has nulls.
Duplicate rows / duplicate OrderIDs	0 / 0 (also 0 duplicate TrackingNumbers)
Date column	All 1,200 values are valid dates (2023-01-01 to 2025-06-30)
Text consistency	No leading/trailing spaces, no mixed casing, no spelling variants
Numeric consistency	No negative or zero values; prices already at 2 decimals

2. Change log

Change ID	Description	Impact	Status
CR001	Filled 309 blank CouponCode values with 'NO_COUPON'. A blank means no coupon was used, so mode imputation (e.g. 'FREESHIP') would have invented discounts.	Preserved 309 records; 0 nulls remain	Resolved
CR002	Checked full-row duplicates and duplicate OrderIDs (drop_duplicates on OrderID, keep first).	0 found, 0 removed; 1,200 records retained	Verified
CR003	Standardised Date to ISO 8601 text (YYYY-MM-DD).	1,200 valid dates; 0 incorrectly formatted	Verified
CR004	Trimmed whitespace; Proper Case on Product, PaymentMethod, OrderStatus, ReferralSource; upper-cased ID and code columns.	0 values needed changing; formats now enforced in script	Verified
CR005	Enforced numeric types: Quantity and ItemsInCart as integers; UnitPrice and TotalPrice rounded and exported with 2 decimals.	0 values changed; TotalPrice = Quantity x UnitPrice on all 1,200 rows	Verified

3. Verification gate (Project 2 threshold)

Test	Result	Status
Duplicate OrderIDs	0 of 1,200	PASS
Incorrectly formatted dates	0 of 1,200	PASS
Remaining null values	0	PASS
TotalPrice differs from Quantity x UnitPrice	0	PASS

4. Observations (flagged, not changed)

These look like business-logic questions rather than formatting errors, so no data was altered. They are listed for stakeholder review.

Ref	Observation	Suggested follow-up
OB1	487 orders with status Pending or Cancelled still carry a TrackingNumber.	Confirm whether tracking IDs are pre-assigned at order creation.
OB2	TotalPrice equals Quantity x UnitPrice even when a coupon (SAVE10, WINTER15, FREESHIP) is present, so discounts are not reflected.	Confirm whether TotalPrice is meant to be pre-discount.
OB3	11 CustomerIDs appear on more than one order.	Expected for repeat customers; no action needed.